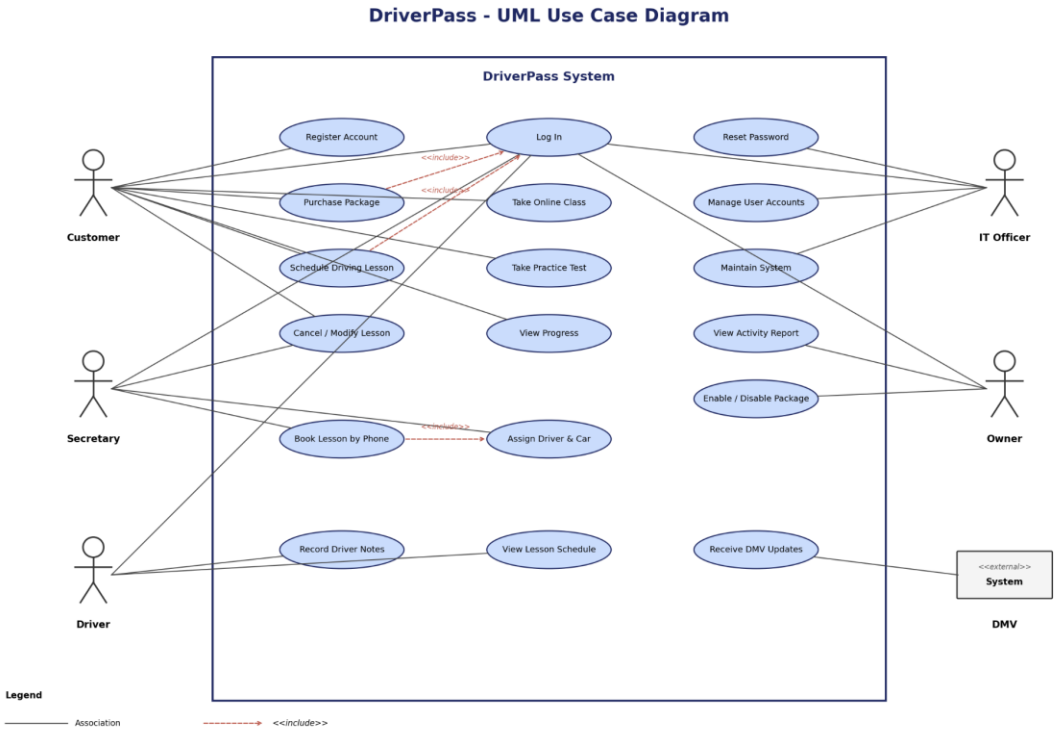
Isaac L. Cabrera Matos

CS-255

DriverPass System Design Document

This report outlines the design for DriverPass according to the requirements of the client. The system will be designed such that customers can book training packages and arrange lessons. Staff members will also be able to manage their work effectively using the system.

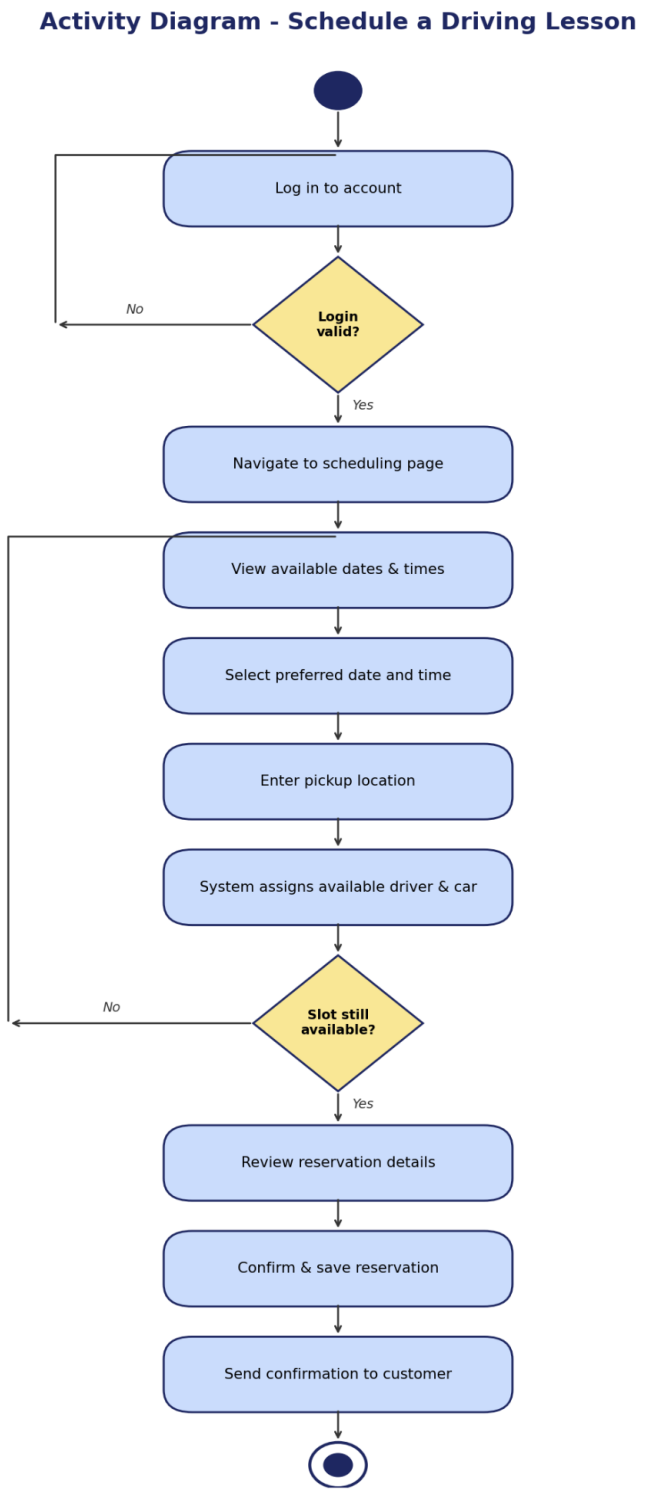
UML Use Case Diagram



Use Case Diagram is shown above based on how the different users use the DriverPass application, which includes Customers, Secretaries, Drivers, Owner, and IT Officer. Customers can carry out actions such as registering for accounts, buying packages, making bookings, cancellations, or modifying classes. On the management side, the staff includes managing bookings and allocation of drivers and cars.

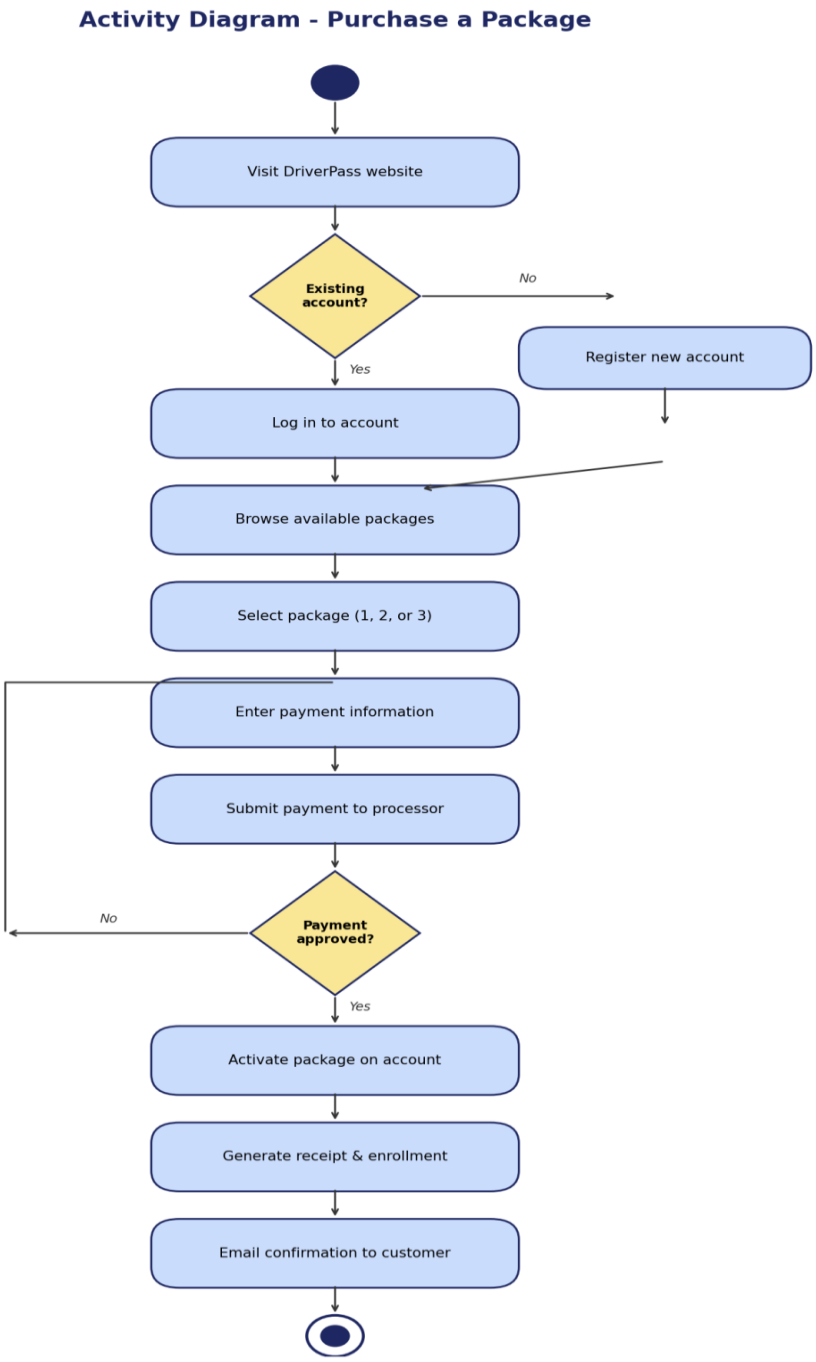
UML Activity Diagrams

Schedule a Driving Lesson



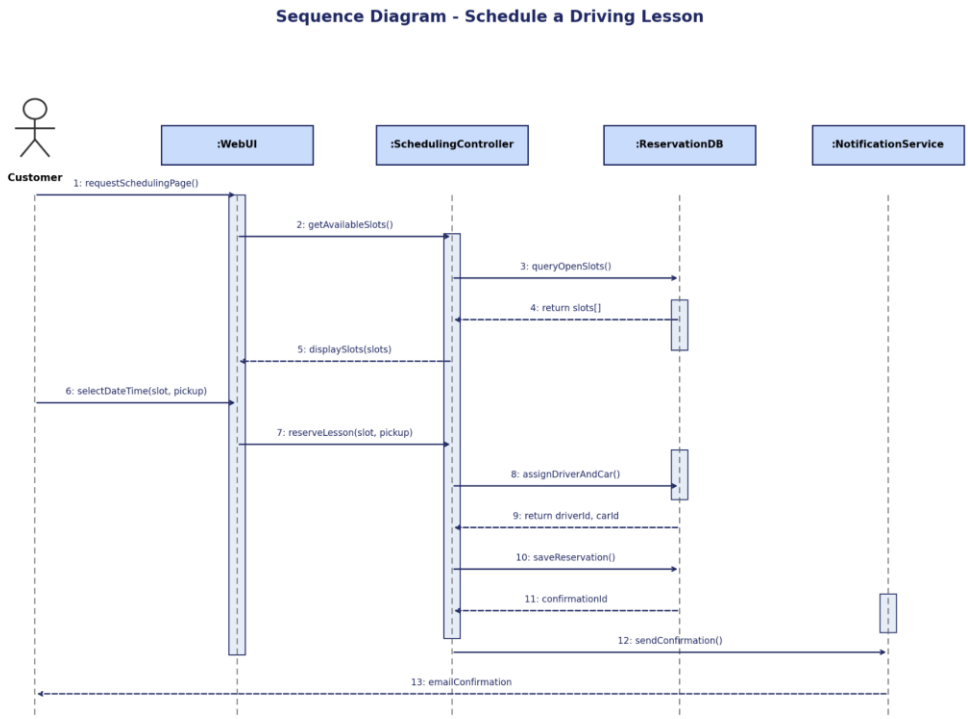
In the above diagram, it shows how the customer logs in, chooses an appropriate time slot, and makes a booking. Each class is allocated a driver, car, and time slot. In addition, the program abides by the client's condition that both pick-up and drop-off points coincide, and each lesson should last for two hours.

Purchase a Package



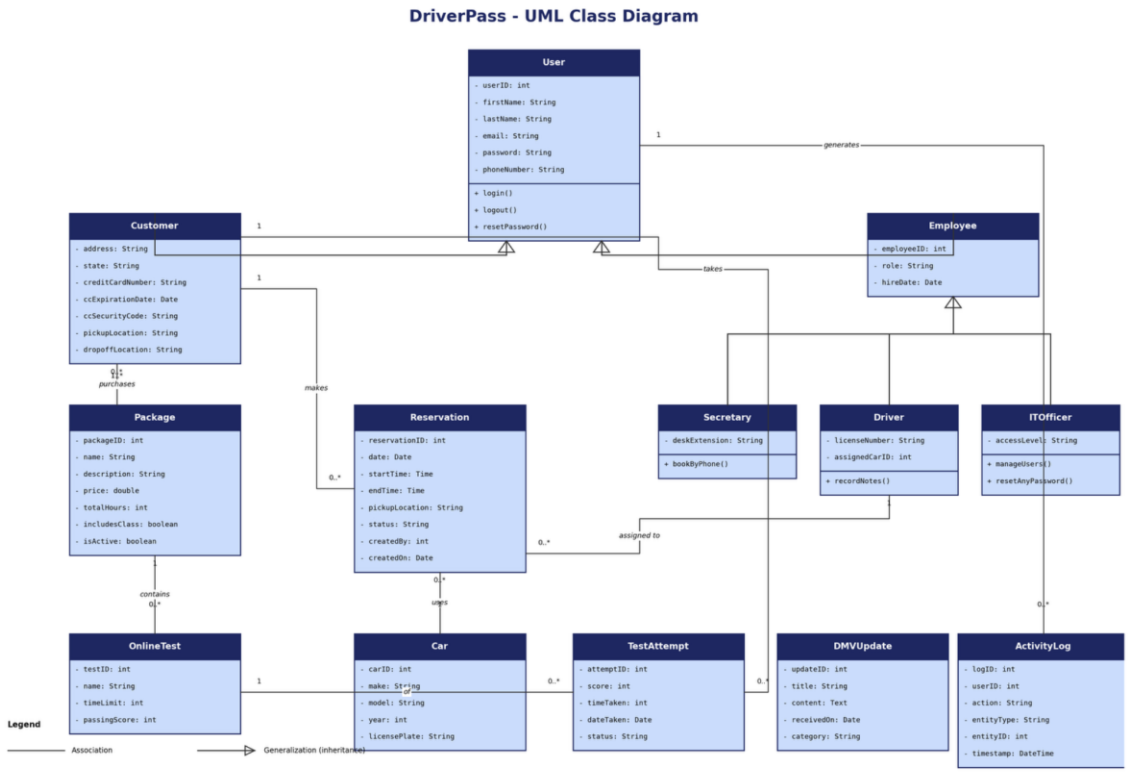
In the above, the client will choose the training package, provide payment details, and make the purchase. This will result in the transaction being confirmed and the customer's account being updated. This fulfills the requirement for a range of packages incorporating lessons and instructions.

UML Sequence Diagram



This diagram shows the procedure used for reserving a lesson. The user will start communication with the system using the user interface. The system will then check whether the database has any available vehicles. Once the vehicle and driver are assigned, the system will document everything and communicate with the user through an acknowledgment message.

UML Class Diagram



The class diagram is the structure of the system. Classes that are used in this system include: customer, driver, vehicle, reservation, package, payment, lesson, and report. There exists an association between the classes, customer with reservation, and package while for driver and vehicle the relationship occurs in the lesson class.

Technical Requirements

The DriverPass system will be deployed using cloud infrastructure, for example, AWS or Microsoft Azure, ensuring scalability, availability, and reliability. The choice between AWS and Azure will be based on factors such as total cost of ownership, existing contracts, integration capabilities, security offerings, and the technical fit with our system requirements. This selection process will help ensure alignment with business goals and stakeholder expectations. The DriverPass system will be accessible with web browsers and mobiles using a responsive front-end based on modern frameworks, for instance, React or Angular.

The back end will be developed based on a framework like Node.js with Express or Java Spring Boot, and it will handle the implementation of a RESTful API, handling system logic and interactions. Relational databases will keep customers, reservations, vehicles, packages, payments, and logs, for instance, MySQL or PostgreSQL.

The system will support authorization mechanisms and role-based access control, allowing different users, such as Customer, Secretary, Driver, Owner, and IT Officer, access depending on their permissions. All traffic will be protected by HTTPS encryption. In addition, enhanced security and monitoring features will be incorporated, such as detailed audit logging of user activities, regular security audits, and the integration of intrusion detection systems to help identify and respond to unauthorized access attempts. These measures ensure compliance with data protection requirements and maintain traceability of all actions within the system.

DMV service integration will enable the system to get updates about rules, policies, and practice tests. Payment processing will be done by PCI compliance payment processor, e.g., Stripe.

DriverPass will use cloud infrastructure capabilities, provide backup and disaster recovery services, and avoid data losses. Activity logging will record all actions performed by any user, and it will allow tracking all modifications made by any person in the system.

The system will be able to allow an offline viewing-only mode, which allows the user to access the reports downloaded by them when they are not connected to the internet, but prevents any changes to the data.

Hardware & Infrastructure: Cloud servers, Application Servers, Database servers, Client machines. Software & Tools: Web browsers, Front End Frameworks, Back End Frameworks, Database Management Systems, Payment Gateway Integration, and Security Tools like Encryption and Access Control.